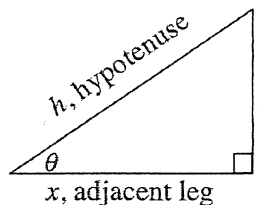


## 1. Solving Right Triangles

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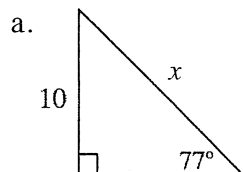
$$\sin(\theta) = \frac{\text{opposite leg}}{\text{hypotenuse}} = \frac{y}{h}$$

$$\cos(\theta) = \frac{\text{adjacent leg}}{\text{hypotenuse}} = \frac{x}{h}$$

$$\tan(\theta) = \frac{\text{opposite leg}}{\text{adjacent leg}} = \frac{y}{x}$$

$$\text{Also, by the Pythagorean Theorem: } h^2 = x^2 + y^2$$

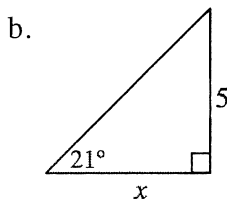
Solution to problem 2-12:



$$\sin(77^\circ) = \frac{10}{x}$$

$$x \sin(77^\circ) = 10$$

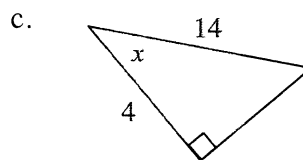
$$x = \frac{10}{\sin(77^\circ)} \approx 10.263$$



$$\tan(21^\circ) = \frac{5}{x}$$

$$x \tan(21^\circ) = 5$$

$$x = \frac{5}{\tan(21^\circ)} \approx 13.025$$

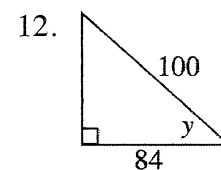
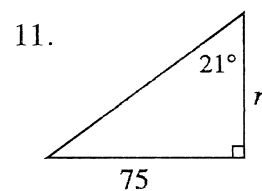
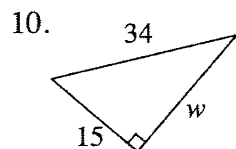
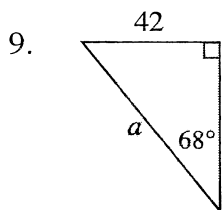
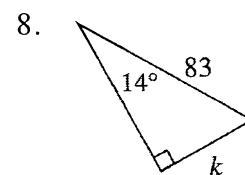
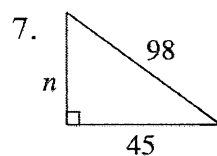
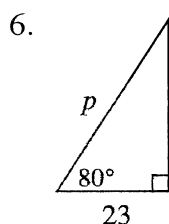
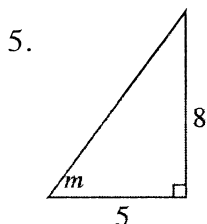
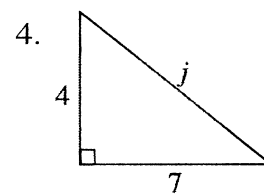
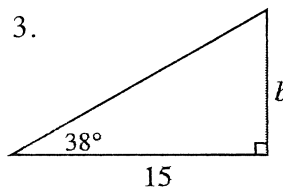
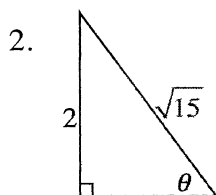
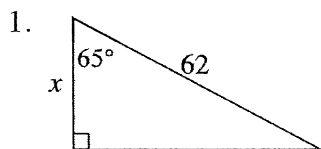


$$\cos(x) = \frac{4}{14}$$

$$\cos^{-1}(\cos(x)) = \cos^{-1}\left(\frac{4}{14}\right)$$

$$x = \cos^{-1}\left(\frac{4}{14}\right) \approx 73.389^\circ$$

Solve for the variable in each of the following triangles.



$$1. \cos(65^\circ) = \frac{x}{62}$$

$$x = 62 \cdot 0.4226$$

$$x = 26.2$$

$$2. \sin(\theta) = \frac{2}{\sqrt{15}}$$

$$\theta = \sin^{-1}(0.5164)$$

$$\theta = 31.1^\circ$$

$$3. \tan(38^\circ) = \frac{b}{15}$$

$$b = 15 \cdot 0.7813$$

$$b = 11.7$$

$$4. 4^2 + 7^2 = j^2$$

$$\sqrt{65} = \sqrt{j^2}$$

$$j = 8.06$$

$$5. \tan(m) = \frac{8}{5} = 1.6$$

$$m = \tan^{-1}(1.6)$$

$$m = 58.0^\circ$$

$$6. \cos(80^\circ) = \frac{23}{p}$$

$$p = \frac{23}{\cos(80^\circ)} = \frac{23}{0.1736}$$

$$p = 132.5$$

$$7. 45^2 + n^2 = 98^2$$

$$n^2 = 9604 - 2025 = 7579$$

$$n = \sqrt{7579} = 87.1$$

$$8. \sin(14^\circ) = \frac{k}{83}$$

$$k = 83 \cdot 0.2419$$

$$k = 20.1$$

$$9. \sin(68^\circ) = \frac{42}{a}$$

$$a = \frac{42}{0.9272}$$

$$a = 45.3$$

$$10. 15^2 + w^2 = 34^2$$

$$w^2 = 1156 - 225 = 931$$

$$w = \sqrt{931} = 30.5$$

$$11. \tan(21^\circ) = \frac{r}{75}$$

$$r = 75 \cdot 0.3839$$

$$r = 28.8$$

$$12. \cos(y) = \frac{84}{100}$$

$$y = \cos^{-1}(0.84)$$

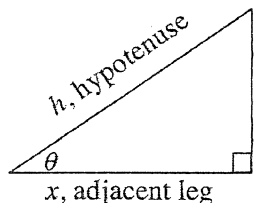
$$y = 32.9^\circ$$

Allen Cook

## 1. Solving Right Triangles

Three trigonometric ratios and the Pythagorean Theorem can be used to solve for the missing side lengths and/or angle measurements in any right triangle.

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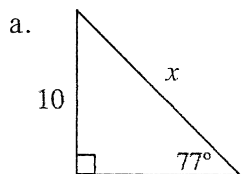
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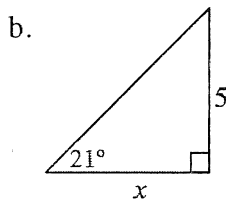
Solution to problem 2-12:



$$\sin(77^\circ) = \frac{10}{x}$$

$$x \sin(77^\circ) = 10$$

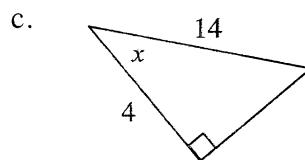
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$$\tan(21^\circ) = \frac{5}{x}$$

$$x \tan(21^\circ) = 5$$

$$x = \frac{5}{\tan(21^\circ)} \approx 13.025$$



$$\cos(x) = \frac{4}{14}$$

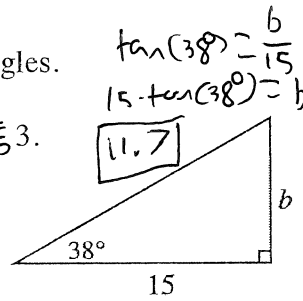
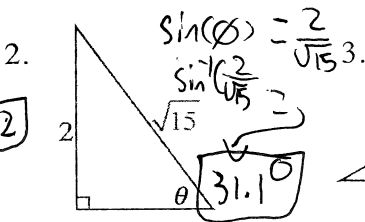
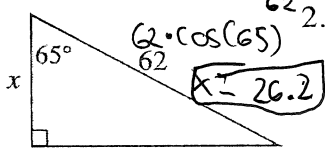
$$\cos^{-1}(\cos(x)) = \cos^{-1}\left(\frac{4}{14}\right)$$

$$x = \cos^{-1}\left(\frac{4}{14}\right) \approx 73.389^\circ$$

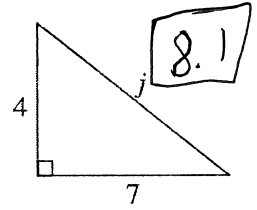
# Alton Cook

Solve for the variable in each of the following triangles.

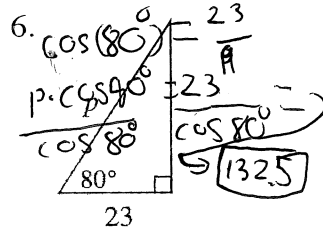
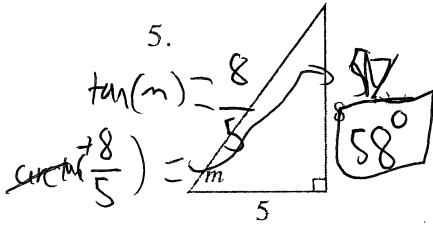
1.



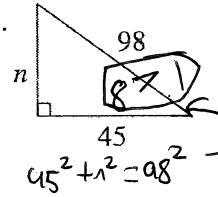
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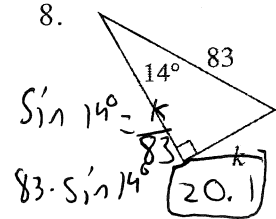
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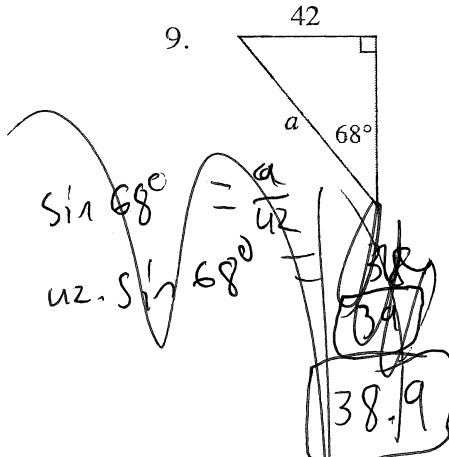
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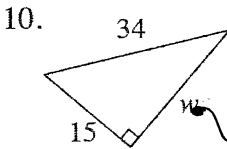
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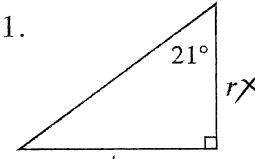
9.



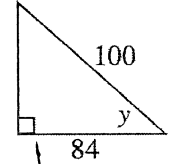
10.



11.



12.



$$a \cdot \sin(68) = \frac{42}{a}$$

$$a \cdot \sin(68) = 42$$

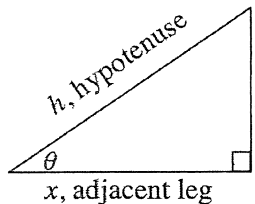
$$\sin(68) = \frac{42}{a}$$

$$a = \frac{42}{\sin(68)} = 45.3$$

# 1. Solving Right Triangles

Three trigonometric ratios and the Pythagorean Theorem can be used to solve for the missing side lengths and/or angle measurements in any right triangle.

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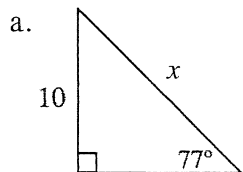
$$\sin(\theta) = \frac{\text{opposite leg}}{\text{hypotenuse}} = \frac{y}{h}$$

$$\cos(\theta) = \frac{\text{adjacent leg}}{\text{hypotenuse}} = \frac{x}{h}$$

$$\tan(\theta) = \frac{\text{opposite leg}}{\text{adjacent leg}} = \frac{y}{x}$$

$$\text{Also, by the Pythagorean Theorem: } h^2 = x^2 + y^2$$

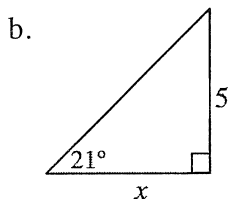
Solution to problem 2-12:



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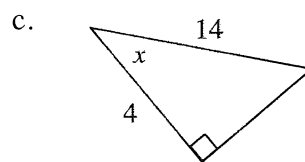
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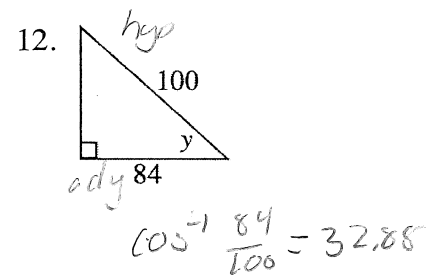
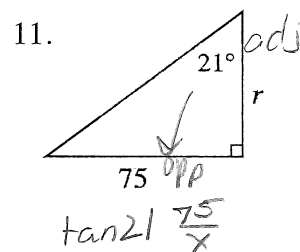
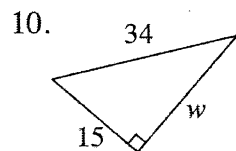
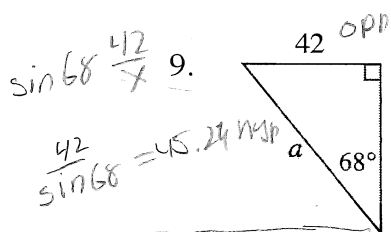
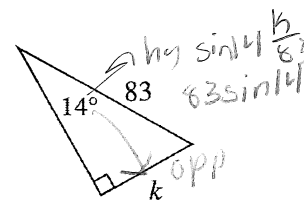
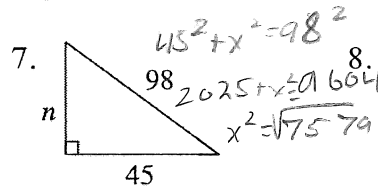
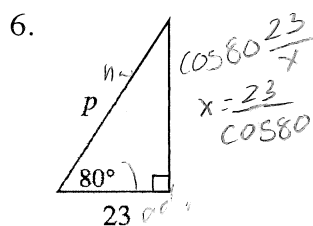
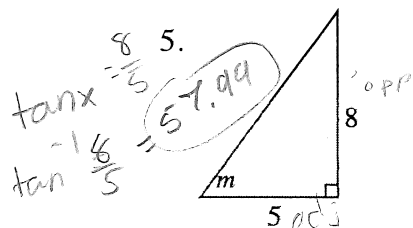
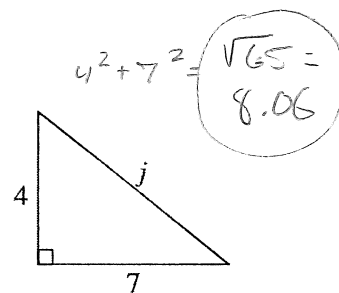
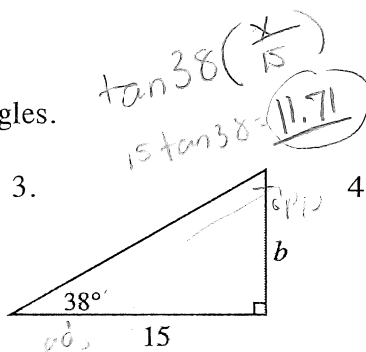
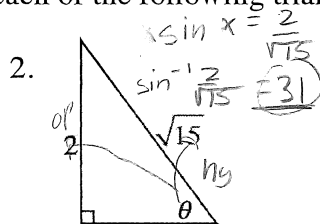
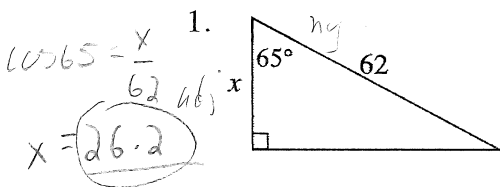
$$\cos(x) = \frac{4}{14}$$

$$\cos^{-1}(\cos(x)) = \cos^{-1}\left(\frac{4}{14}\right)$$

$$x = \cos^{-1}\left(\frac{4}{14}\right) \approx 73.389^\circ$$

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Solve for the variable in each of the following triangles.



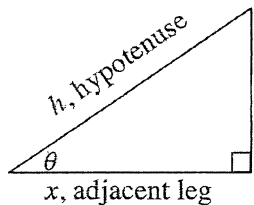
1. 26.2
2. 31.09
3. 11.71
4. 8.06
5. 57.99
6. 132.45
7. 87.05
8. 20.68
9. 45.29
10. 30.51
11. 195.38
12. 32.85

Joselyn P.

## 1. Solving Right Triangles

Three trigonometric ratios and the Pythagorean Theorem can be used to solve for the missing side lengths and/or angle measurements in any right triangle.

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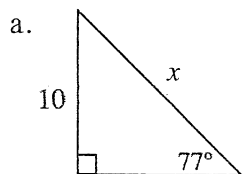
$$\sin(\theta) = \frac{\text{opposite leg}}{\text{hypotenuse}} = \frac{y}{h}$$

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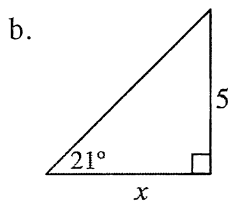
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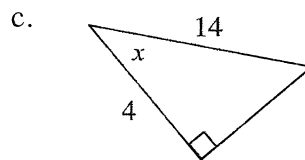
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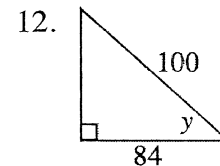
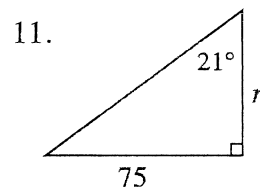
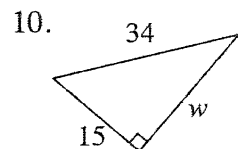
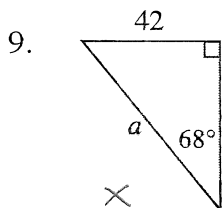
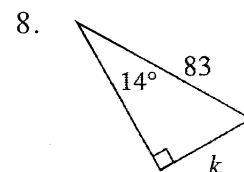
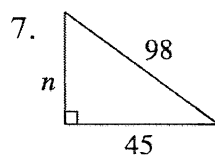
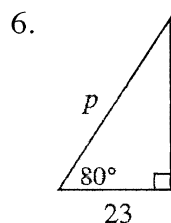
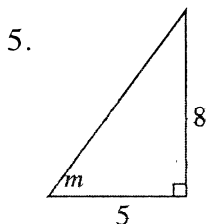
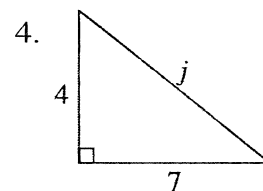
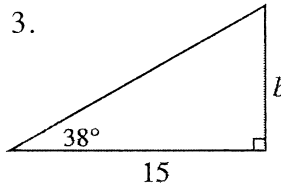
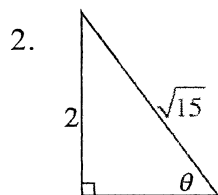
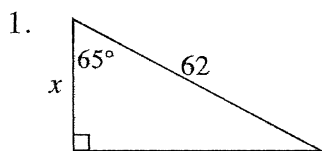
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$$x = \cos^{-1}\left(\frac{4}{14}\right) \approx 73.389^\circ$$

# SOH CAH TOA

Solve for the variable in each of the following triangles.



$$\begin{aligned} 1. \cos(65) &= \frac{x}{62} \\ (62)\cos(65) &= x \\ 26.2 &\approx x \end{aligned}$$

$$\begin{aligned} 2. \sin(\theta) &= \frac{2}{\sqrt{15}} \\ \theta &= \sin^{-1}\left(\frac{2}{\sqrt{15}}\right) \\ \theta &\approx 31.09 \end{aligned}$$

$$\begin{aligned} 3. \tan(38) &= \frac{b}{15} \\ 15\tan(38) &= b \\ 11.7 &\approx b \end{aligned}$$

$$\begin{aligned} 4. \quad 4^2 + 7^2 &= j^2 \\ 16 + 49 &= j^2 \\ 65 &= j^2 \\ \sqrt{65} &= j \end{aligned}$$

$$\begin{aligned} 5. \tan(m) &= \frac{8}{5} \\ m &= \tan^{-1}\left(\frac{8}{5}\right) \\ m &\approx 57.995 \end{aligned}$$

$$\begin{aligned} 6. \cos(80) &= \frac{23}{x} \\ x\cos(80) &= 23 \\ x &= \frac{23}{\cos(80)} \\ x &\approx 132.45 \end{aligned}$$

$$\begin{aligned} 7. \quad 45^2 + x^2 &= 98^2 \\ 2025 + x^2 &= 9604 \\ x^2 &= 7579 \\ x &\approx 87.06 \end{aligned}$$

$$\begin{aligned} 8. \sin(14) &= \frac{k}{83} \\ 83\sin(14) &= k \\ 20.08 &\approx k \end{aligned}$$

$$\begin{aligned} 9. \sin(68) &= \frac{42}{a} \\ a\sin(68) &= 42 \\ a &= \frac{42}{\sin(68)} \\ a &\approx 45.3 \end{aligned}$$

$$\begin{aligned} 10. \quad 15^2 + w^2 &= 34^2 \\ 225 + w^2 &= 1156 \\ w^2 &= 931 \\ w &= \sqrt{931} \end{aligned}$$

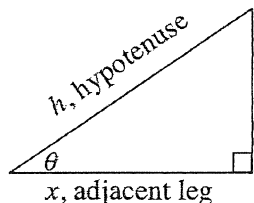
$$\begin{aligned} 11. \tan(21) &= \frac{75}{r} \\ r\tan(21) &= 75 \\ r &= \frac{75}{\tan(21)} \\ r &\approx 195.38 \end{aligned}$$

$$\begin{aligned} 12. \cos y &= \frac{84}{100} \\ y &= \cos^{-1}\left(\frac{84}{100}\right) \\ y &\approx 32.86 \end{aligned}$$

## 1. Solving Right Triangles

Three trigonometric ratios and the Pythagorean Theorem can be used to solve for the missing side lengths and/or angle measurements in any right triangle.

In the triangle below, when the sides are described relative to the angle  $\theta$ , the opposite leg is  $y$  and the adjacent leg is  $x$ . The hypotenuse is  $h$  regardless of which acute angle is used.



$$\sin(\theta) = \frac{\text{opposite leg}}{\text{hypotenuse}} = \frac{y}{h}$$

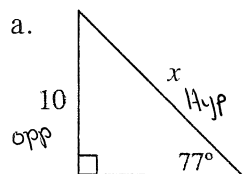
$$\cos(\theta) = \frac{\text{adjacent leg}}{\text{hypotenuse}} = \frac{x}{h}$$

$$\tan(\theta) = \frac{\text{opposite leg}}{\text{adjacent leg}} = \frac{y}{x}$$

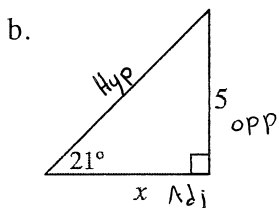
Also, by the Pythagorean Theorem:  $h^2 = x^2 + y^2$

Solution to problem 2-12:

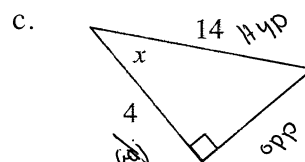
*SOH CAH TOA*



$$\begin{aligned}\sin(77^\circ) &= \frac{10}{x} \\ x \sin(77^\circ) &= 10 \\ x &= \frac{10}{\sin(77^\circ)} \approx 10.263\end{aligned}$$



$$\begin{aligned}\tan(21^\circ) &= \frac{5}{x} \\ x \tan(21^\circ) &= 5 \\ x &= \frac{5}{\tan(21^\circ)} \approx 13.025\end{aligned}$$



$$\begin{aligned}\cos(x) &= \frac{4}{14} \\ \cos^{-1}(\cos(x)) &= \cos^{-1}\left(\frac{4}{14}\right) \\ x &= \cos^{-1}\left(\frac{4}{14}\right) \approx 73.389^\circ\end{aligned}$$

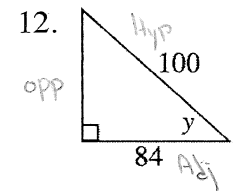
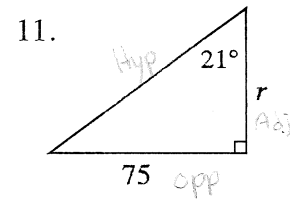
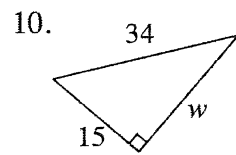
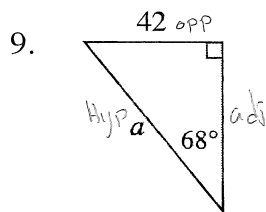
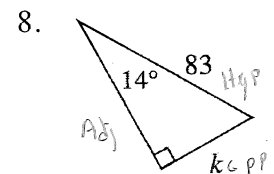
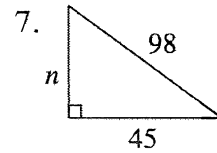
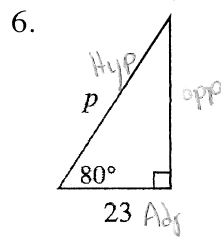
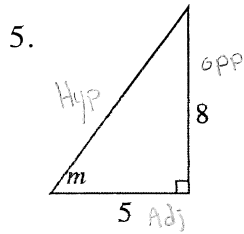
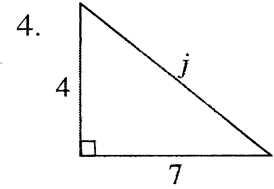
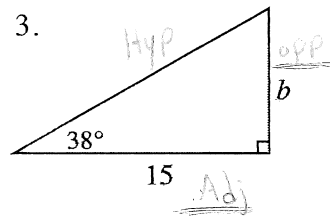
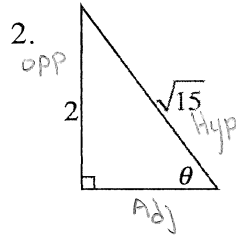
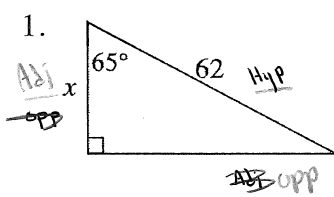
*To find an angle you take the inverse.*

*Degrees*

$$\begin{aligned}x \sin 77^\circ &= \frac{10}{x} \cdot x \\ \cancel{x} \sin 77^\circ &= \frac{10}{\cancel{x}} \\ \sin 77^\circ &= \frac{10}{x} \\ x &= \frac{10}{\sin 77^\circ}\end{aligned}$$

*$x \tan 21^\circ = \frac{5}{x}$*

Solve for the variable in each of the following triangles.



$$1. \sin 65^\circ = \frac{x}{62} \cdot 62$$

$$62 \cos 65^\circ = x$$

$$x = 26.202$$

$$2. \sin \theta = \frac{2}{\sqrt{15}}$$

$$x = \sin^{-1} \frac{2}{\sqrt{15}}$$

$$x = 31.09$$

$$3. \tan 38^\circ = \frac{x}{15} \cdot 15$$

$$15 \tan 38^\circ = x$$

$$x = 11.72$$

$$4. 4^2 + 7^2 = x^2$$

$$16 + 49 = 65$$

$$\sqrt{65} = 8.1$$

$$5. \tan(x) = \frac{4}{5}$$

$$x = \tan^{-1} \left( \frac{4}{5} \right)$$

$$x = 57.99$$

$$6. \cos 80^\circ = \frac{23}{x}$$

$$x \cos 80^\circ = 23$$

$$\frac{x \cos 80^\circ}{\cos 80^\circ} = \frac{23}{\cos 80^\circ}$$

$$x = 132.45$$

$$7. 98^2 - 45^2 = x^2$$

$$9604 - 2025 = 7579$$

$$\sqrt{7579} = 87.057$$

$$8. \sin 14^\circ = \frac{x}{83} \cdot 83$$

$$83 \sin 14^\circ = x$$

$$x = 20.08$$

$$9. \sin 68^\circ = \frac{42}{x} \cdot x$$

$$x \sin 68^\circ = 42$$

$$\frac{x \sin 68^\circ}{\sin 68^\circ} = \frac{42}{\sin 68^\circ}$$

$$x = 45.298$$

$$10. 34^2 - 15^2 = x^2$$

$$1156 - 225 = 931$$

$$\sqrt{931} = 30.51$$

$$11. \tan(21^\circ) = \frac{25}{x} \cdot x$$

$$x \tan 21^\circ = 75$$

$$\frac{x \tan 21^\circ}{\tan 21^\circ} = \frac{75}{\tan 21^\circ}$$

$$x = 195.4$$

$$12. \cos x = \frac{84}{100}$$

$$x = \cos^{-1} \left( \frac{84}{100} \right)$$

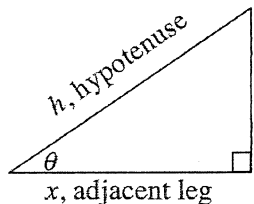
$$x = 32.86$$

Arlilyn Stetten

## 1. Solving Right Triangles

Three trigonometric ratios and the Pythagorean Theorem can be used to solve for the missing side lengths and/or angle measurements in any right triangle.

In the triangle below, when the sides are described relative to the angle  $\theta$ , the opposite leg is  $y$  and the adjacent leg is  $x$ . The hypotenuse is  $h$  regardless of which acute angle is used.



$$\sin(\theta) = \frac{\text{opposite leg}}{\text{hypotenuse}} = \frac{y}{h}$$

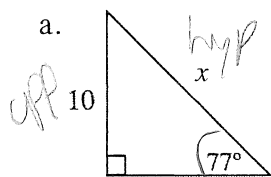
$$\cos(\theta) = \frac{\text{adjacent leg}}{\text{hypotenuse}} = \frac{x}{h}$$

$$\tan(\theta) = \frac{\text{opposite leg}}{\text{adjacent leg}} = \frac{y}{x}$$

Also, by the Pythagorean Theorem:  $h^2 = x^2 + y^2$

SOH CAH TOA

Solution to problem 2-12:



$$\sin(77^\circ) = \frac{10}{x}$$

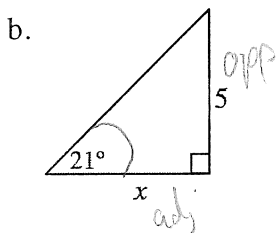
$$x \sin(77^\circ) = 10$$

$$x = \frac{10}{\sin(77^\circ)} \approx 10.263$$

$$x \sin 77^\circ = \frac{10}{\cancel{x} \cdot \cancel{x}} \cdot x$$

$$\frac{x \sin 77^\circ}{\cancel{\sin 77^\circ}} = \frac{10}{\cancel{\sin 77^\circ}}$$

$$x = \frac{10}{\sin 77^\circ}$$



$$\tan(21^\circ) = \frac{5}{x}$$

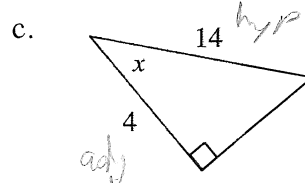
$$x \tan(21^\circ) = 5$$

$$x = \frac{5}{\tan(21^\circ)} \approx 13.025$$

$$\tan 21^\circ = \frac{5}{x}$$

$$\frac{x \tan 21^\circ}{\cancel{\tan 21^\circ}} = \frac{5}{\cancel{\tan 21^\circ}}$$

$$x = \frac{5}{\tan 21^\circ}$$



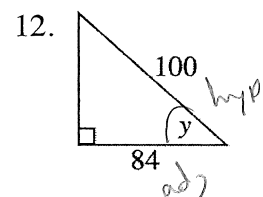
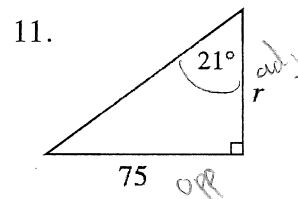
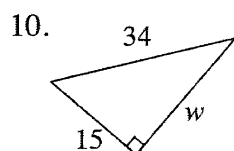
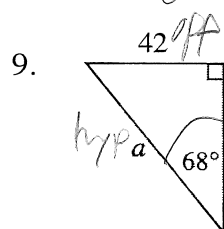
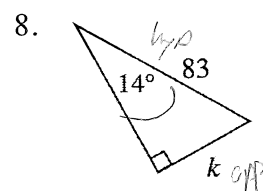
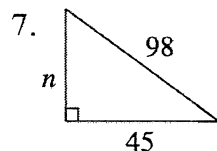
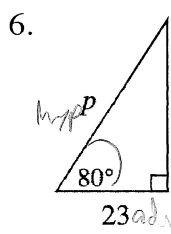
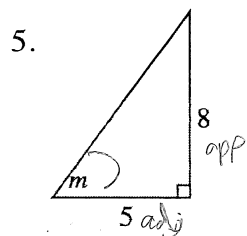
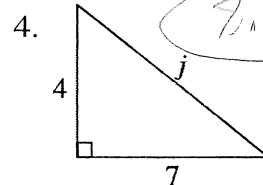
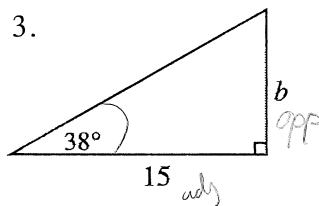
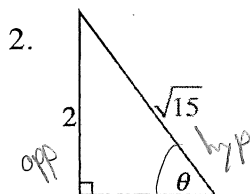
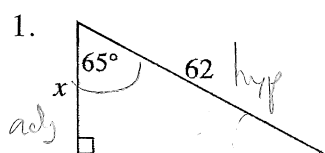
$$\cos(x) = \frac{4}{14}$$

$$\cos^{-1}(\cos(x)) = \cos^{-1}\left(\frac{4}{14}\right)$$

$$x = \cos^{-1}\left(\frac{4}{14}\right) \approx 73.389^\circ$$

to find angle, take the inverse  
 $\cos x = \frac{4}{14}$   
 $x = \cos^{-1}\left(\frac{4}{14}\right)$

Solve for the variable in each of the following triangles.



cos CAh  
 $\cos 65 = \frac{x}{62}$   
 $(62) \cos 65 = x$   
 $x = 26.2$

$\sin(\theta) = \frac{2}{\sqrt{15}}$   
 $\theta = \sin^{-1}\left(\frac{2}{\sqrt{15}}\right)$   
 $\theta = 31.09$

tan  
 $\tan 38 = \frac{b}{15}$   
 $b = (15) \tan 38$   
 $b = 11.72$

4.  $\sin 14 = \frac{k}{83}$   
 $83 \sin 14 = k$   
 $k = 20.08$

tan  
 $\tan(m) = \frac{8}{5}$   
 $m = \tan^{-1}\left(\frac{8}{5}\right)$   
 $m = 57.99^\circ$

cos  
 $\cos 80 = \frac{23}{p}$   
 $p \cos 80 = 23$   
 $p = \frac{23}{\cos 80}$   
 $p = 132.45$

$98^2 - 45^2$   
 $a = 87.06$

5.  $\sin 68 = \frac{42}{a}$   
 $a \sin 68 = 42$   
 $a = \frac{42}{\sin 68}$   
 $a = 45.3$

6.  $34^2 - 15^2$   
 $w = \sqrt{109}$   
 $30.61$

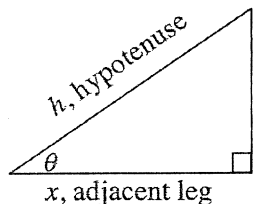
7.  $\tan 21 = \frac{75}{r}$   
 $r \tan 21 = 75$   
 $r = \frac{75}{\tan 21}$   
 $r = 195.38$

8.  $\cos CAH$   
 $\cos(y) = \frac{84}{100}$   
 $y = \cos^{-1}\left(\frac{84}{100}\right)$   
 $y = 32.86$

# 1. Solving Right Triangles

Three trigonometric ratios and the Pythagorean Theorem can be used to solve for the missing side lengths and/or angle measurements in any right triangle.

In the triangle below, when the sides are described relative to the angle  $\theta$ , the opposite leg is  $y$  and the adjacent leg is  $x$ . The hypotenuse is  $h$  regardless of which acute angle is used.



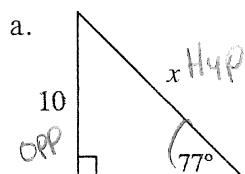
$$\sin(\theta) = \frac{\text{opposite leg}}{\text{hypotenuse}} = \frac{y}{h}$$

$$\cos(\theta) = \frac{\text{adjacent leg}}{\text{hypotenuse}} = \frac{x}{h}$$

$$\tan(\theta) = \frac{\text{opposite leg}}{\text{adjacent leg}} = \frac{y}{x}$$

Also, by the Pythagorean Theorem:  $h^2 = x^2 + y^2$

Solution to problem 2-12:



$$\sin(77^\circ) = \frac{10}{x}$$

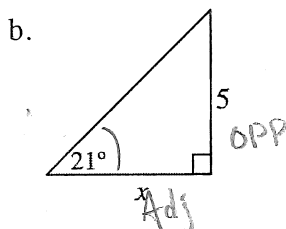
$$x \sin(77^\circ) = 10$$

$$x = \frac{10}{\sin(77^\circ)} \approx 10.263$$

$$x \sin 77 = \frac{10}{x} \cdot x = 10$$

$$\frac{x \sin 77}{\sin 77} = \frac{10}{\sin 77}$$

$$x = 10.263$$



$$\tan(21^\circ) = \frac{5}{x}$$

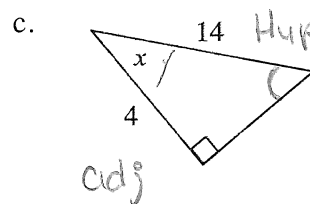
$$x \tan(21^\circ) = 5$$

$$x = \frac{5}{\tan(21^\circ)} \approx 13.025$$

$$\tan 21 = \frac{5}{x} \cdot x$$

$$\frac{x \tan 21}{\tan 21} = \frac{5}{\tan 21}$$

$$x \approx 13.025$$



$$\cos(x) = \frac{4}{14}$$

$$\cos^{-1}(\cos(x)) = \cos^{-1}\left(\frac{4}{14}\right)$$

$$x = \cos^{-1}\left(\frac{4}{14}\right) \approx 73.389^\circ$$

to find angle  
take the  
inverse

$$\cos x = \frac{4}{14}$$

$$x = \cos^{-1}\left(\frac{4}{14}\right)$$

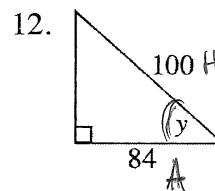
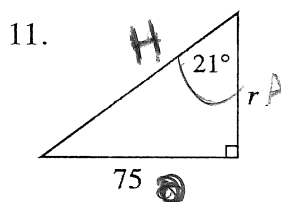
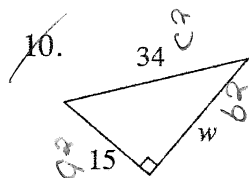
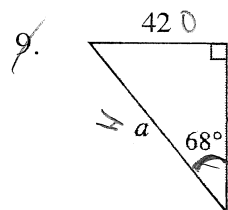
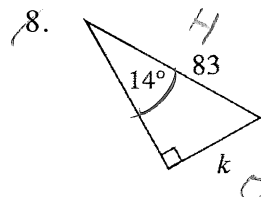
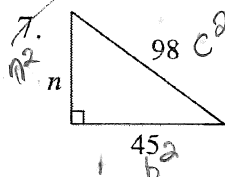
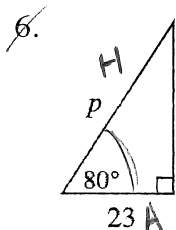
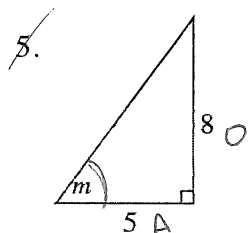
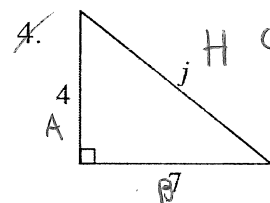
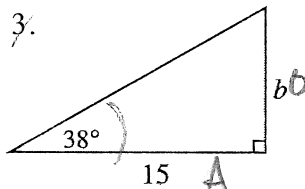
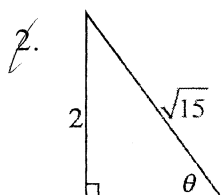
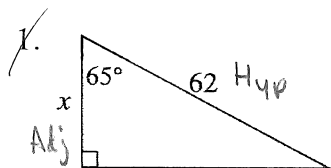
all in calc  
at once

21780/6400

$$h^2 = x^2 + y^2$$

SOH CAH TOA

Solve for the variable in each of the following triangles.



1.)  $\cos 65 = \frac{x}{62} \cdot 62 = x$

2.)  $\sin^{-1}\left(\frac{2}{\sqrt{15}}\right) = \theta$

3.)  $\tan 38 = \frac{b}{15} \cdot 15 = b$

$$62 \cos 65 = x$$

$$x \approx 26.2$$

$$\theta = 31.09$$

$$15 \tan 38 = b$$

$$b \approx 11.72$$

4.)  $4^2 + 7^2 = j^2$

$$16 + 49 = j^2$$

$$\sqrt{65} = \sqrt{j^2}$$

$$8.06 \approx j$$

5.)  $\tan^{-1}\left(\frac{8}{3}\right)$

$$m \approx 57.99$$

6.)  $\cos 80 = \frac{23}{p}$

$$\frac{23}{\cos 80} = p$$

$$132.45 \approx p$$

7.)  $98^2 - 45^2 = 7579$

$$\sqrt{7579}$$

$$\downarrow$$

$$n = 87.05$$

$$n \approx 1.06$$

8.)  $\sin 14 = \frac{k}{83} \cdot 83 = k$

$$83 \sin 14 = k$$

$$20.07 = k$$

9.)  $\frac{42}{\sin 68} = \frac{42}{a}$

$$\downarrow \quad \downarrow$$

$$45.30 \approx a$$

10.)  $c^2 - a^2 = b^2$

$$c^2 - a^2$$

$$\downarrow \quad \downarrow$$

$$34^2 - 15^2 = 931$$

$$\sqrt{931} = \sqrt{b^2}$$

$$30.51 = b$$

11.)  $\frac{75}{\cos 21} = \frac{75}{r}$

$$\downarrow \quad \downarrow$$

$$70.01 = r$$

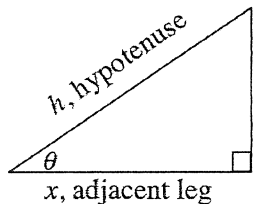
12.)  $\cos^{-1}\left(\frac{84}{100}\right) = y$

$$32.85 = y$$

# 1. Solving Right Triangles

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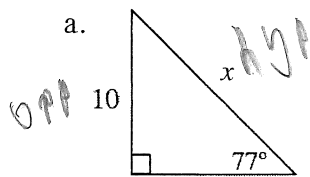
$$\sin(\theta) = \frac{\text{opposite leg}}{\text{hypotenuse}} = \frac{y}{h}$$

$$\cos(\theta) = \frac{\text{adjacent leg}}{\text{hypotenuse}} = \frac{x}{h}$$

$$\tan(\theta) = \frac{\text{opposite leg}}{\text{adjacent leg}} = \frac{y}{x}$$

Also, by the Pythagorean Theorem:  $h^2 = x^2 + y^2$

Solution to problem 2-12:



$$\sin(77^\circ) = \frac{10}{x}$$

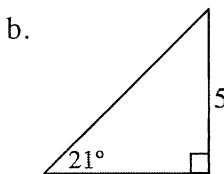
$$x \sin(77^\circ) = 10$$

$$x = \frac{10}{\sin(77^\circ)} \approx 10.263$$

$$\sin 77^\circ = \frac{10}{x} \cdot x$$

$$\frac{x \sin 77^\circ}{\sin 77^\circ} = \frac{10}{\sin 77^\circ}$$

$$x = 10.263$$



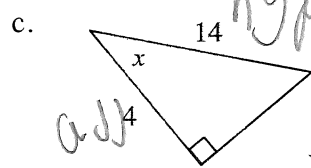
$$\tan(21^\circ) = \frac{5}{x}$$

$$x \tan(21^\circ) = 5$$

$$x = \frac{5}{\tan(21^\circ)} \approx 13.025$$

$$\tan 21^\circ = \frac{5}{x}$$

$$\frac{x \tan 21^\circ}{\tan 21^\circ} = \frac{5}{\tan 21^\circ}$$



$$\cos(x) = \frac{4}{14}$$

$$\cos^{-1}(\cos(x)) = \cos^{-1}\left(\frac{4}{14}\right)$$

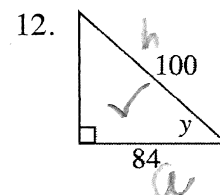
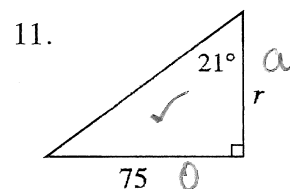
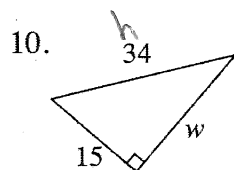
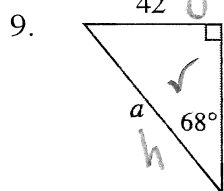
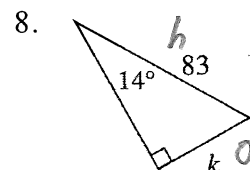
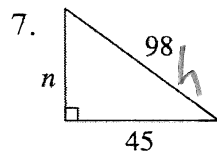
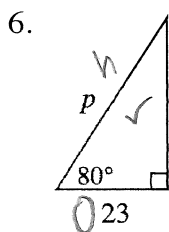
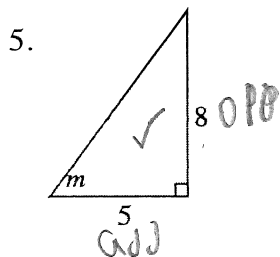
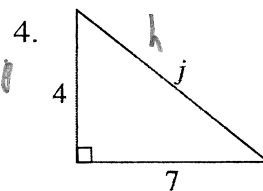
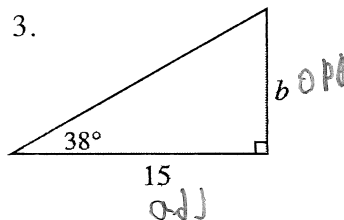
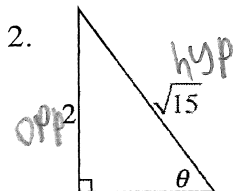
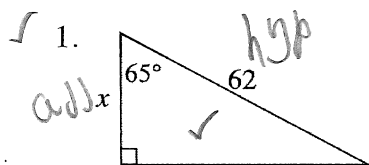
$$x = \cos^{-1}\left(\frac{4}{14}\right) \approx 73.389^\circ$$

to find  
an angle  
take the  
inverse

$$\cos x = \frac{4}{14}$$

$$x = \cos^{-1}\left(\frac{4}{14}\right)$$

Solve for the variable in each of the following triangles.



①  $\cos 65 = \frac{62}{x}$   
 $x \cos 65 = 62$   
 $\frac{\cos 65}{\cos 65} \frac{\cos 65}{\cos 65}$   
 $x = 146.7$

②

③  $\tan 38 = \frac{b}{15}$   
 $15 \cdot \tan 38 = b$   
 $b = 11.719$

④

⑤  $\tan m = \frac{8}{5}$   
 $m = \tan^{-1}\left(\frac{8}{5}\right)$   
 $m = 57.994$

⑥  $\sin 80 = \frac{23}{p}$   
 $p \sin 80 = 23$   
 $p = \frac{23}{\sin 80}$   
 $p = 23.354$

⑦

⑧

⑨  $\sin 68 = \frac{42}{a}$   
 $a \sin 68 = 42$   
 $a = \frac{42}{\sin 68}$   
 $a = 45.298$

⑩

⑪  $\sin 21 = \frac{75}{r}$   
 $r \sin 21 = 75$   
 $r = \frac{75}{\sin 21}$   
 $r = 209.282$

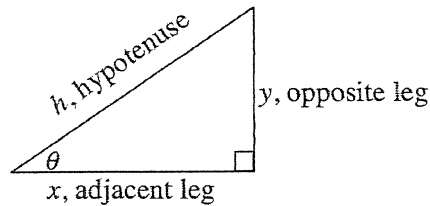
⑫  $\cos y = \frac{84}{100}$   
 $y = \cos^{-1}\left(\frac{84}{100}\right)$   
 $y = 32.859$

Caleb B

## 1. Solving Right Triangles

Three trigonometric ratios and the Pythagorean Theorem can be used to solve for the missing side lengths and/or angle measurements in any right triangle.

In the triangle below, when the sides are described relative to the angle  $\theta$ , the opposite leg is  $y$  and the adjacent leg is  $x$ . The hypotenuse is  $h$  regardless of which acute angle is used.



$$\sin(\theta) = \frac{\text{opposite leg}}{\text{hypotenuse}} = \frac{y}{h}$$

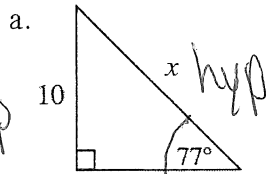
$$\cos(\theta) = \frac{\text{adjacent leg}}{\text{hypotenuse}} = \frac{x}{h}$$

$$\tan(\theta) = \frac{\text{opposite leg}}{\text{adjacent leg}} = \frac{y}{x}$$

$$\text{Also, by the Pythagorean Theorem: } h^2 = x^2 + y^2$$

Solution to problem 2-12:

SOH CAH TOA



$$\sin(77^\circ) = \frac{10}{x}$$

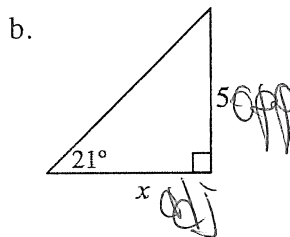
$$x \sin(77^\circ) = 10$$

$$x = \frac{10}{\sin(77^\circ)} \approx 10.263$$

$$x \sin 77^\circ = \frac{10}{\cancel{\sin 77^\circ}} \cdot \cancel{\sin 77^\circ} \cdot x$$

$$x \sin 77^\circ = \frac{10}{\sin 77^\circ}$$

$$x = 10.263$$

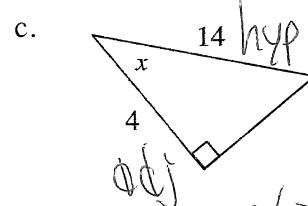


$$\tan(21^\circ) = \frac{5}{x}$$

$$x \tan(21^\circ) = 5$$

$$x = \frac{5}{\tan(21^\circ)} \approx 13.025$$

$$\tan 21^\circ = \frac{5}{x}$$



$$\cos(x) = \frac{4}{14}$$

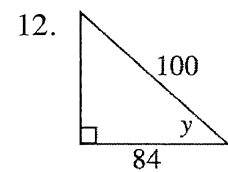
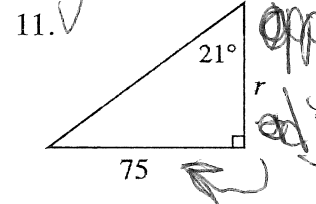
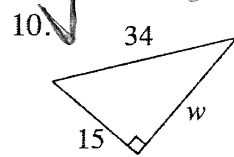
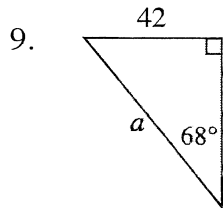
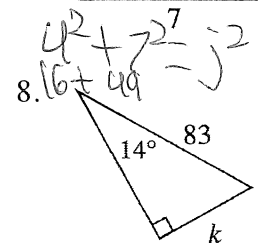
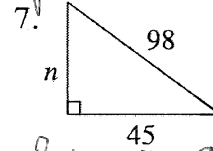
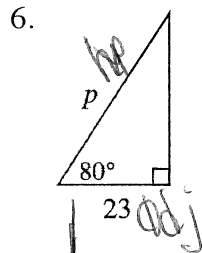
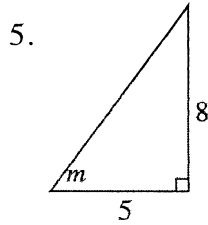
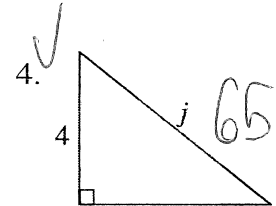
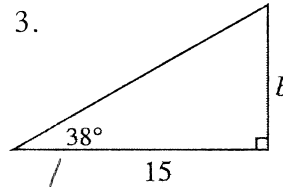
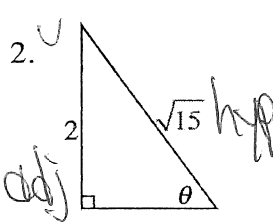
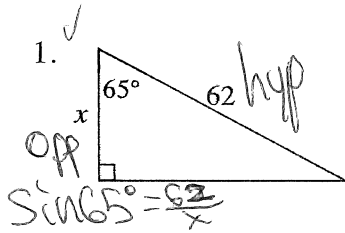
$$\cos^{-1}(\cos(x)) = \cos^{-1}\left(\frac{4}{14}\right)$$

$$x = \cos^{-1}\left(\frac{4}{14}\right) \approx 73.389^\circ$$

$$\cos x = \frac{4}{14}$$

$$x = \cos^{-1}\left(\frac{4}{14}\right)$$

Solve for the variable in each of the following triangles.



$15^2 + w^2 = 34^2$   
 $225 + w^2 = 1156$   
 $w^2 = 931$   
 $w = \sqrt{931} \approx 30.51$

1.  $x \sin 65^\circ = 62$

2.  $\cos \theta = \frac{2}{\sqrt{15}}$

$11 \times \tan 21^\circ = 75$

$x = \frac{62}{\sin 65^\circ} = 68.409$

$\theta = \cos^{-1}\left(\frac{2}{\sqrt{15}}\right) = 58.909^\circ$

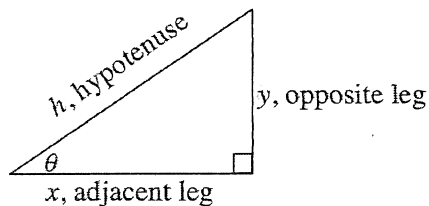
$x = \frac{75}{\tan 21^\circ} = 195.38$



## 1. Solving Right Triangles

Three trigonometric ratios and the Pythagorean Theorem can be used to solve for the missing side lengths and/or angle measurements in any right triangle.

In the triangle below, when the sides are described relative to the angle  $\theta$ , the opposite leg is  $y$  and the adjacent leg is  $x$ . The hypotenuse is  $h$  regardless of which acute angle is used.



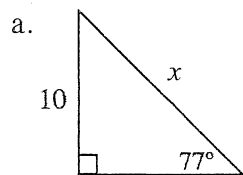
$$\sin(\theta) = \frac{\text{opposite leg}}{\text{hypotenuse}} = \frac{y}{h}$$

$$\cos(\theta) = \frac{\text{adjacent leg}}{\text{hypotenuse}} = \frac{x}{h}$$

$$\tan(\theta) = \frac{\text{opposite leg}}{\text{adjacent leg}} = \frac{y}{x}$$

Also, by the Pythagorean Theorem:  $h^2 = x^2 + y^2$

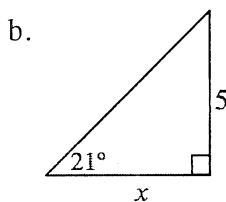
Solution to problem 2-12:



$$\sin(77^\circ) = \frac{10}{x}$$

$$x \sin(77^\circ) = 10$$

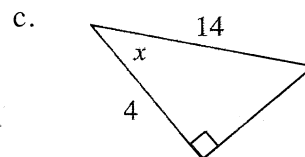
$$x = \frac{10}{\sin(77^\circ)} \approx 10.263$$



$$\tan(21^\circ) = \frac{5}{x}$$

$$x \tan(21^\circ) = 5$$

$$x = \frac{5}{\tan(21^\circ)} \approx 13.025$$



$$\cos(x) = \frac{4}{14}$$

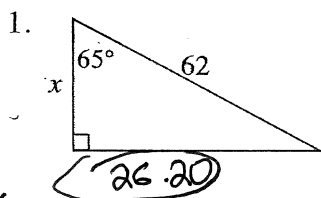
$$\cos^{-1}(\cos(x)) = \cos^{-1}\left(\frac{4}{14}\right)$$

$$x = \cos^{-1}\left(\frac{4}{14}\right) \approx 73.389^\circ$$

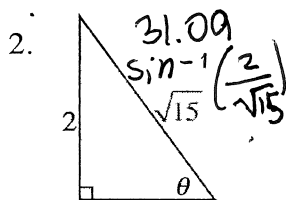
# SOH CAH ~~ATO~~A

Solve for the variable in each of the following triangles.

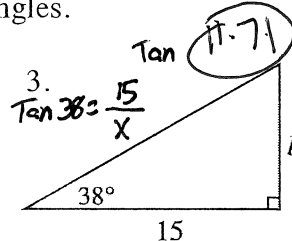
1.  $\cos 65 = \frac{x}{62}$   
 $62 \cos 65 = x$



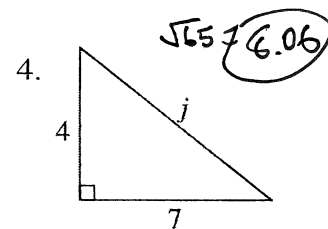
$26.20$



$31.09$   
 $\sin^{-1}(\frac{2}{\sqrt{15}})$

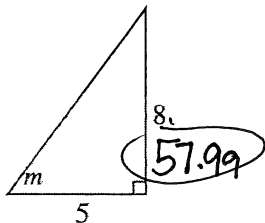


$\tan 38 = \frac{15}{b}$   
 $11.71$

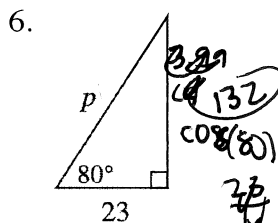


$5.657$   
 $6.06$

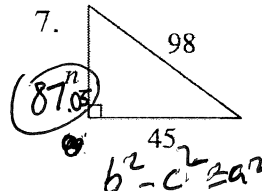
5.  $\tan x = \frac{8}{5}$   
 $\tan^{-1} \frac{8}{5}$



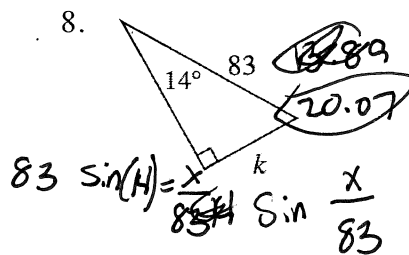
$57.99$



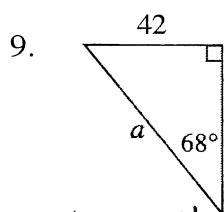
$3.21$   
 $132$   
 $\cos(80)$   
 $23$



$87.08$   
 $b^2 - c^2 = a^2$

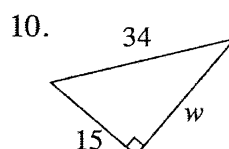


$83 \sin(14) = \frac{x}{83}$   
 $20.07$   
 $\sin \frac{x}{83}$

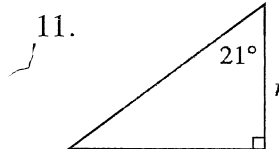


$\sin(68) = \frac{42}{a}$   
 $45.29$   
 $x = \frac{\sin(68)}{42}$

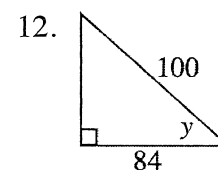
$42$   
 $\sin(68)$



$30.31$   
 $15$



$\tan(21) = \frac{75}{r}$   
 $195.3$   
 $75$   
 $\tan(21)$

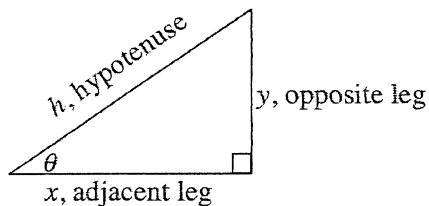


$\cos^{-1} \frac{84}{100} = 32.85$

## 1. Solving Right Triangles

Three trigonometric ratios and the Pythagorean Theorem can be used to solve for the missing side lengths and/or angle measurements in any right triangle.

In the triangle below, when the sides are described relative to the angle  $\theta$ , the opposite leg is  $y$  and the adjacent leg is  $x$ . The hypotenuse is  $h$  regardless of which acute angle is used.



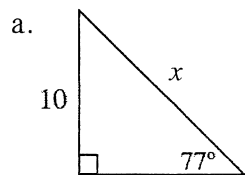
$$\sin(\theta) = \frac{\text{opposite leg}}{\text{hypotenuse}} = \frac{y}{h}$$

$$\cos(\theta) = \frac{\text{adjacent leg}}{\text{hypotenuse}} = \frac{x}{h}$$

$$\tan(\theta) = \frac{\text{opposite leg}}{\text{adjacent leg}} = \frac{y}{x}$$

Also, by the Pythagorean Theorem:  $h^2 = x^2 + y^2$

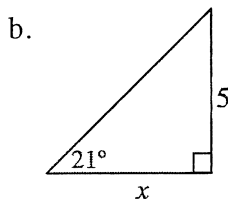
Solution to problem 2-12:



$$\sin(77^\circ) = \frac{10}{x}$$

$$x \sin(77^\circ) = 10$$

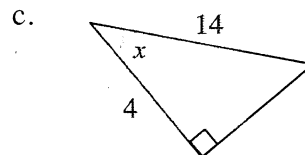
$$x = \frac{10}{\sin(77^\circ)} \approx 10.263$$



$$\tan(21^\circ) = \frac{5}{x}$$

$$x \tan(21^\circ) = 5$$

$$x = \frac{5}{\tan(21^\circ)} \approx 13.025$$



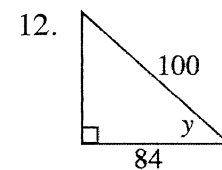
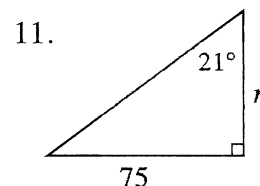
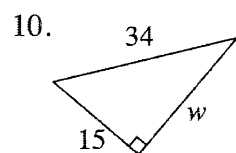
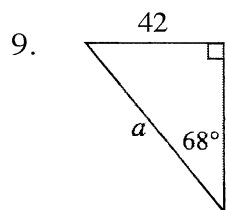
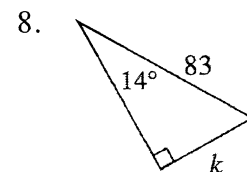
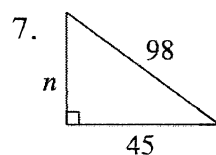
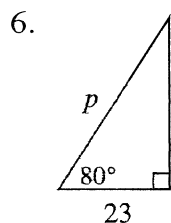
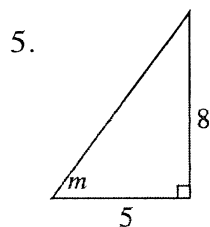
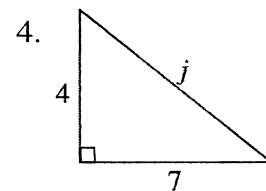
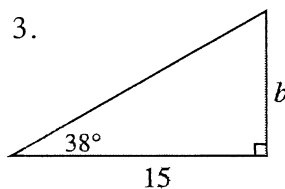
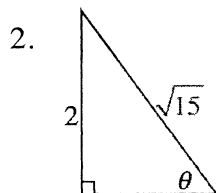
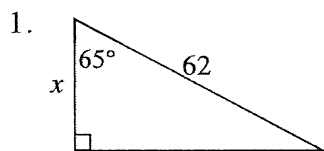
$$\cos(x) = \frac{4}{14}$$

$$\cos^{-1}(\cos(x)) = \cos^{-1}\left(\frac{4}{14}\right)$$

$$x = \cos^{-1}\left(\frac{4}{14}\right) \approx 73.389^\circ$$

# Dalton

Solve for the variable in each of the following triangles.



1.  $\cos 65 = \frac{x}{62}$   
 $62 \cos 65$

2.  $\sin^{-1}(\frac{2}{\sqrt{15}})$   
 $31.09^\circ$

3.  $\tan 38 = \frac{b}{15}$   
 $11.71$

4.  $\sqrt{4^2 + 7^2}$   
 $8.06$

5.  $\tan^{-1}(\frac{8}{5})$   
 $57.99^\circ$

6.  $\cos 80 = \frac{23}{p}$   
 $p = \frac{23}{\cos 80}$   
 $132.9^\circ$

7.  $98^2 = \sqrt{45^2 + n^2}$   
 $87.05$

8.  $83 \sin 14 = k$   
 $20.07$

9.  $\sin 68 = \frac{42}{a}$   
 $a = \frac{42}{\sin 68}$   
 $45.29$

10.  $34^2 = \sqrt{15^2 + w^2}$   
 $30.5$

11.  $\tan 21 = \frac{75}{r}$   
 $r = \frac{75}{\tan 21}$   
 $195.38$

12.  $\cos^{-1}(\frac{84}{100})$   
 $32.85^\circ$